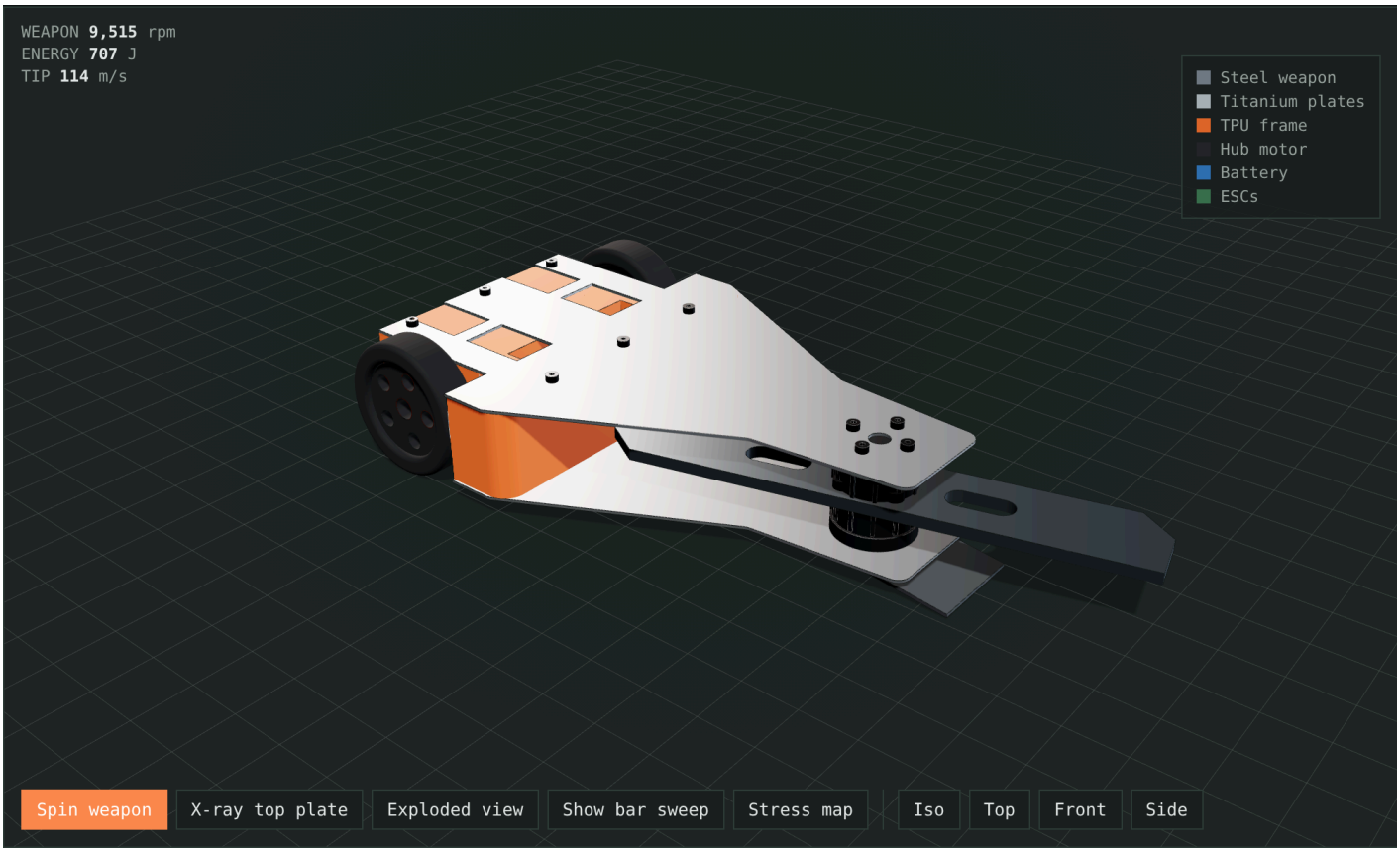
BUILDER PACKAGE · REV A · NOVEMBER 2026

Unskinny Mk II

3 lb (1361 g) beetleweight combat robot · horizontal bar spinner · invertible
Designer: Mahak · github.com/mahak867/unskinny



Mass	1,328 g (limit 1,361 g)	**Weapon**	229 × 38 × 5 mm AR500 bar, 304 g
Weapon speed / energy	~11,000 rpm / ~940 J (4S)	**Spin-up to 90%**	2.0 s
Drive	2× 24 mm brushless planetary 27:1, 2.7 m/s	**Power**	4S 650 mAh LiPo
Structure	1.2 mm Ti-6Al-4V plates, TPU 95A frame	**Footprint**	260 × 164 × 62 mm (wheels)

What's in this package: design upgrades (§1), files list (§2), bought parts (§3), made parts (§4), fasteners and torques (§5), part drawings (§6), wiring (§7), assembly (§8), radio and speed-controller setup (§9), testing (§10), spares and tools (§11), safety (§12).
Which file wins: where a dimension on a drawing and the STEP/DXF files disagree, the STEP/DXF files are correct.

1 · Design upgrades: making it the best bot

These come from the simulations and design checks, ordered by how much they help. P1 do before building · P2 strongly recommended · P3 later improvement.

#	Upgrade	Why (evidence)	Cost
P1	**Lower the weapon to 10–15 mm above the floor.** Mount the motor offset downward and make the bottom plate a short skid under the bar.	The bar currently spins at the robot's mid-height, about 31 mm off the floor. Wedges, forks and wheels sit lower than that, so the bar skims over them. Hitting low is what horizontal spinners win with. Trade-off: it no longer works upside down, so add a quick self-righting method or accept the risk.	Redesign plates and frame; ~0 g
P1	**Use a single-tooth bar** (one impact tooth plus a counterweight lobe).	Bite doubles from 8.2 mm to about 16 mm at 3 m/s, about the same as fully engaging the tooth every hit. In the fight simulator, hits under 10 mm bite deliver only 40–70% of the energy.	Same mass; balance carefully
P1	**Thicken the frame walls at the ESC bays** from about 7 mm to 12 mm, or move the bays toward the middle.	Drawing UMK2-004 shows the ESC bays leave thin walls right behind the wheel slots, the most likely place for a horizontal spinner to hit.	+8 g
P2	**Use replaceable hardened teeth**: two bolt-on S7 inserts (HRC 54–56) on an AR500 bar.	The teeth take all the impact. Swap them between fights instead of the whole 304 g bar.	+10 g
P2	**Use an 850 mAh battery.**	The simulation uses 7.4 of the 7.7 Wh available per match, which is too tight.	+25 g (margin is 33 g)
P2	**Add a second fork set** (longer and lower) for wedge opponents, swappable in under 1 minute.	In the fight simulator, a steel wedge beats Mk II 17 times out of 20 on damage.	Spare part only
P2	**Weapon braking or reverse**: enable braking in the weapon ESC on a switch.	The gyroscopic tilt is about 11× gravity in fast turns. Slowing the weapon before tight manoeuvres keeps the wheels on the floor.	Setting only
P3	**ELRS receiver with telemetry** (weapon rpm, battery voltage).	Lets you check the real rpm against the simulation and catch voltage sag.	~0 g
P3	**Shock-mount the electronics** in 3 mm EVA foam and strap the battery with Velcro over foam.	A 940 J hit sends shock loads through the TPU frame into the ESCs.	+5 g
P3	**Spend the remaining weight on the bar tips** (thicker tips or longer bar).	Stored energy grows with mass × radius². 20 g at the tips adds about 8% energy.	uses margin

Update `params.py` after each change and run `make`. The CAD, simulations and 12 design checks re-run and flag any clash or weight problem.

2 · Files list

File	Use
`out/unskinny_mk2_assembly.step`	Full assembly, opens in any CAD program
`out/bar.dxf`, `out/top_plate.dxf`, `out/bottom_plate.dxf`	Laser or waterjet cut files (mm, 1:1)
`out/bar.step`, `out/top_plate.step`, `out/bottom_plate.step`, `out/frame.step`	3D models of each part
`out/frame.stl`, `out/fork.stl`	3D print / bend reference
`builder/drawings/UMK2-00x_*.svg/png`	Dimensioned part drawings (also in §6)
`builder/drawings/UMK2-E01_wiring.svg`	Wiring diagram (§7)
`builder/drawings/UMK2-A01_exploded_assembly.png`	Exploded assembly with numbered parts (§8)
`unskinny-mk2.html`	Interactive 3D model and simulators, works offline

3 · Bought parts

Item	Qty	Part	Example / source	Spec to match	g
6	1 (+1)	Weapon hub motor 2812	Repeat Robotics 2812 Battle Ready Beetle Hubmotor (palmbeachbots.com)	M4 mount on Ø25 PCD, M5 bell holes on Ø30 PCD, ~900 kV	175
7	2 (+1)	Drive motor	Repeat Robotics Max V2 24 mm, 27:1	Ø24 can, 4S capable	60 ea
11	1 (+1)	Weapon ESC	AM32 or BLHeli_32, 60–80 A	4S, ramp/soft start, brake option	35
12	1 (+1)	Dual drive ESC	Repeat or DualSky combat dual ESC	brushless, reversing, 4S, 5 V BEC	15
13	1	Receiver	ELRS (recommended) / FrSky / FlySky	≥3 channels, failsafe "no pulses"	5
14	1	Power link / switch	Fingertech switch or XT30 removable link	reachable when fully assembled	10
8	3	LiPo battery	GNB / CNHL / Tattu 4S 650 (or 850) mAh 75C	≤ 72 × 36 × 26 mm, XT30	85 ea
5	2 (+2)	Wheel + hub	Fingertech Viper or cast urethane 62 mm	hub bore to match drive output	35 ea
—	1 set	Wire & connectors	14 AWG + 18 AWG silicone, 26 AWG servo leads, XT30 pairs, 3.5 mm bullets, heat-shrink		40
—	1	Transmitter	Any that binds to the receiver (ELRS: RadioMaster)	failsafe + mixing	
—	1	Charger + LiPo bag	balance charger, 4S capable		

4 · Made parts

DWG	Part	Material	Process	Qty	g
UMK2-001	Weapon bar	AR500, 5 mm	Waterjet/laser → grind bevel → balance	2	304.5
UMK2-002	Top plate	Ti-6Al-4V, 1.2 mm	Laser/waterjet → deburr	2	127.6
UMK2-003	Bottom plate	Ti-6Al-4V, 1.2 mm	Laser/waterjet → deburr → drill fork holes	2	127.6
UMK2-004	Frame	TPU 95A	FDM, 5 walls, 45% gyroid, 12× M3 heat-set inserts	2	166.0
UMK2-005	Fork	Ti-6Al-4V, 1.5 mm	Laser → brake bend ≈ 9.5°	3	17.0

Suggested suppliers: SendCutSend or OSH Cut (AR500 and titanium cutting), any local FDM printer or JLCPCB 3D printing for TPU.

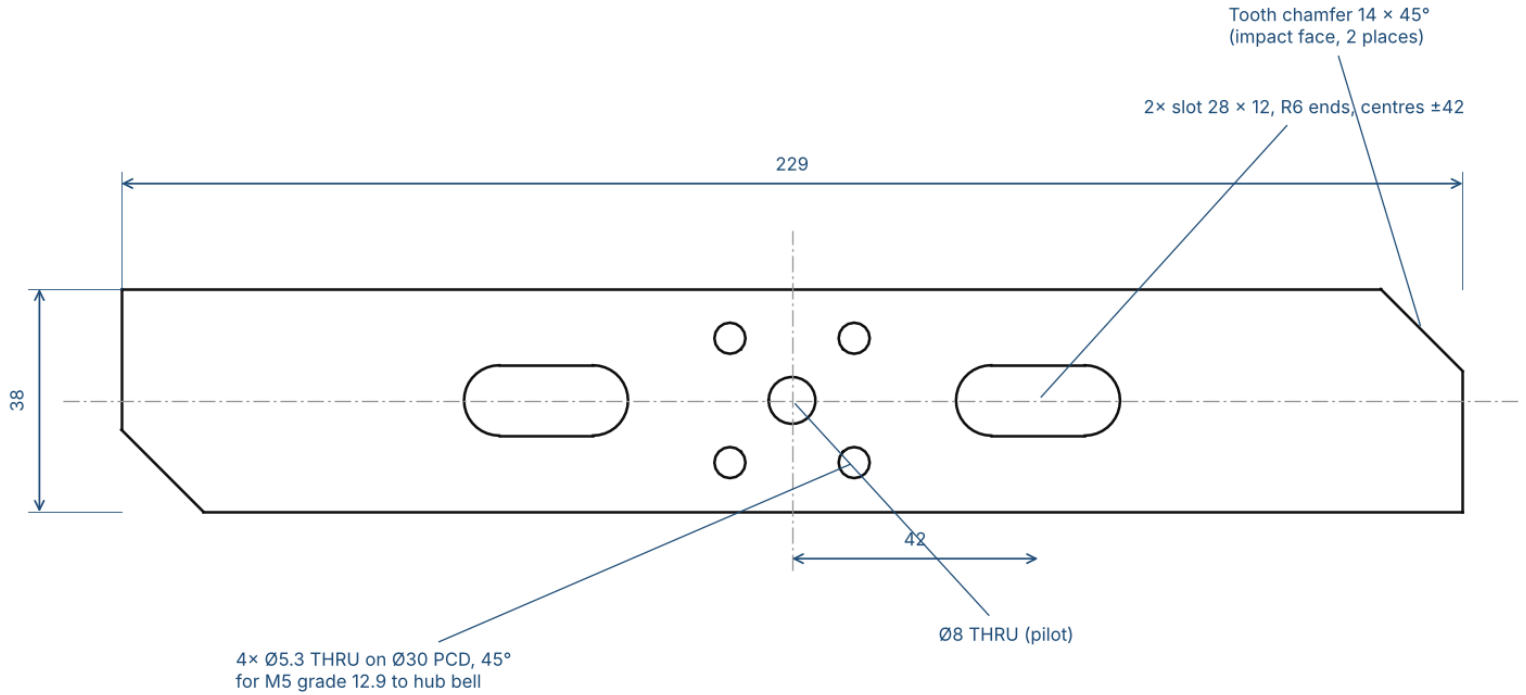
5 · Fasteners and torques

Item	Qty	Fastener	Joins	Torque	Threadlocker
10	4 (+4)	M5 × 12 socket head cap screw, grade 12.9	Bar → hub bell	4.5 N·m	Loctite 243 (blue)
9a	8	M4 × 8 button head, grade 12.9	Plates → hub stator (4 top, 4 bottom)	2.0 N·m	Loctite 243
9b	12	M3 × 8 button head, A2 or 12.9	Plates → frame heat-set inserts	0.5 N·m	none (TPU)
9c	2	M3 × 6 button head + nyloc nut	Fork → bottom plate	0.8 N·m	nyloc
—	12	M3 heat-set inserts, 5.7 mm long	Into frame top and bottom	—	—

Check the screw lengths against the actual motor thread depth before ordering. Screws must not reach the motor windings.

6 · Part drawings

The next 5 landscape sheets: UMK2-001 weapon bar, UMK2-002 top plate, UMK2-003 bottom plate, UMK2-004 frame, UMK2-005 fork.



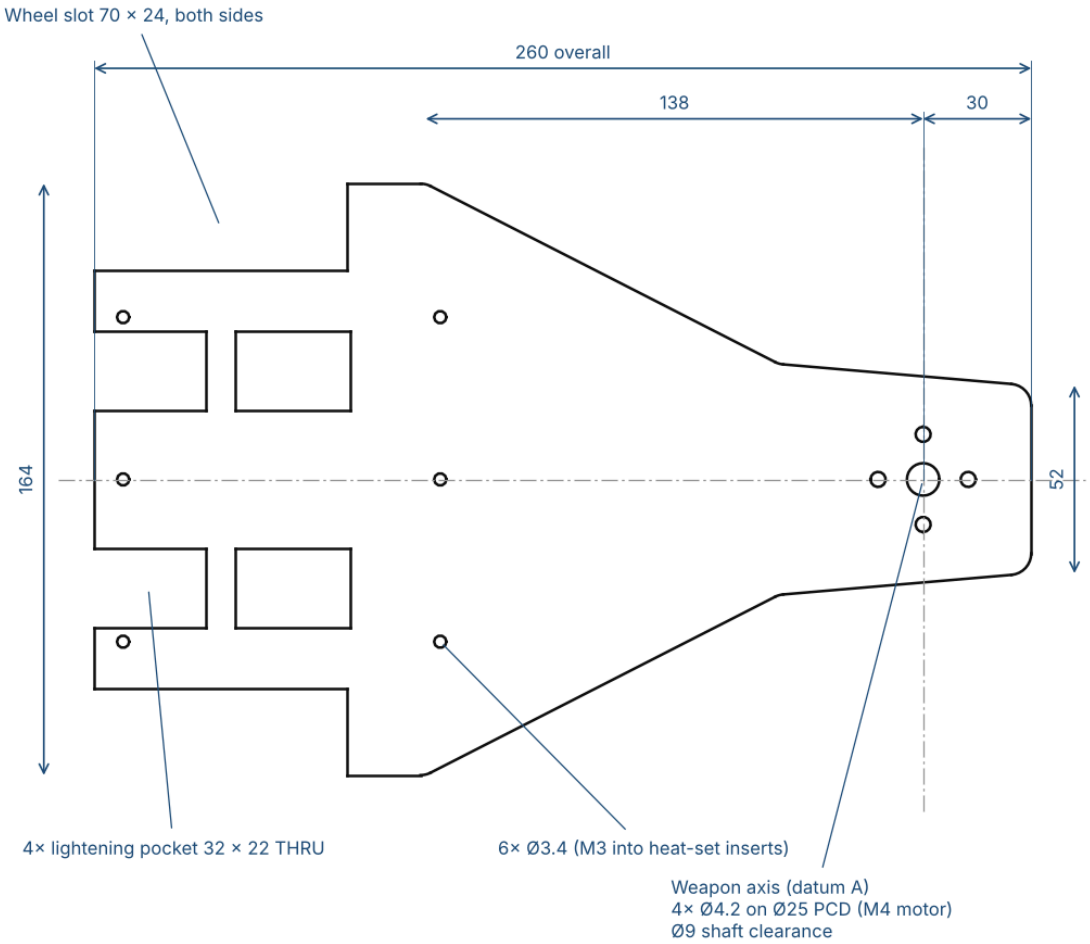
Direction of rotation: counter-clockwise viewed from top. Leading edges = chamfered corners.

PART	DWG NO.	MATERIAL	THICKNESS	QTY	UNITS / SCALE
Weapon bar	**UMK2-001**	**AR500 (or S7 tool steel)**	**5.0 mm**	**2 (1 spare)**	**mm / fit**

PROCESS: Waterjet or laser from bar.dxf. Grind 30° bevel on impact faces. Static + dynamic balance to < 0.5 g·mm.

NOTES: General tol ±0.2; bolt holes +0.1/-0; bore concentric to PCD within 0.05. Break all edges 0.3. Mass 304 g ± 5 g.

UNSKINNY MK II · REV A · 2026-11-07 · Drawn: Mahak · Master geometry: STEP/DXF in out/

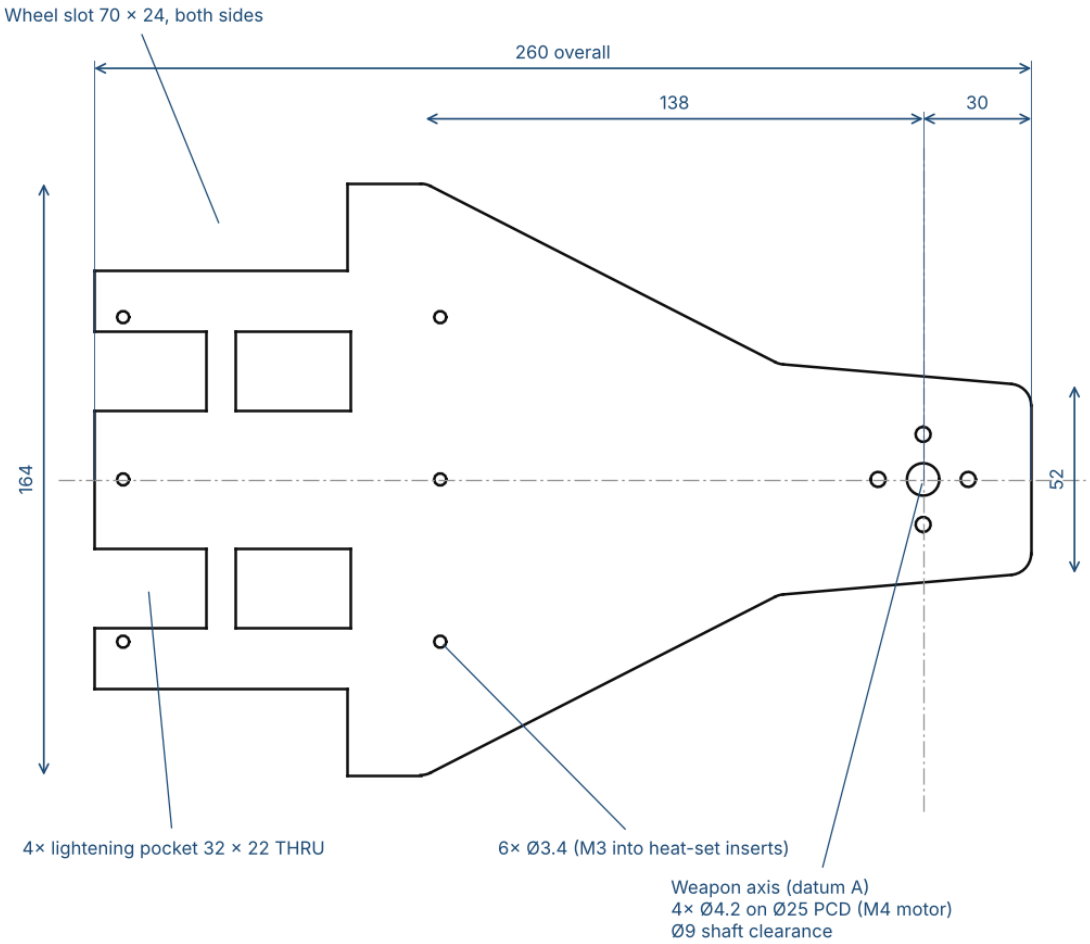


PART	DWG NO.	MATERIAL	THICKNESS	QTY	UNITS / SCALE
Top plate	**UMK2-002**	**Ti-6Al-4V (grade 5)**	**1.2 mm**	**1 (+1 spare)**	**mm / fit**

PROCESS: Laser or waterjet from top_plate.dxf. Deburr. Motor holes reamed for location.

NOTES: Tol ±0.15; corner radii R6; motor PCD true position Ø0.1 to datum A. Bottom plate also carries the fork (drill 2× Ø3.4 per fork drawing).

UNSKINNY MK II · REV A · 2026-11-07 · Drawn: Mahak · Master geometry: STEP/DXF in out/

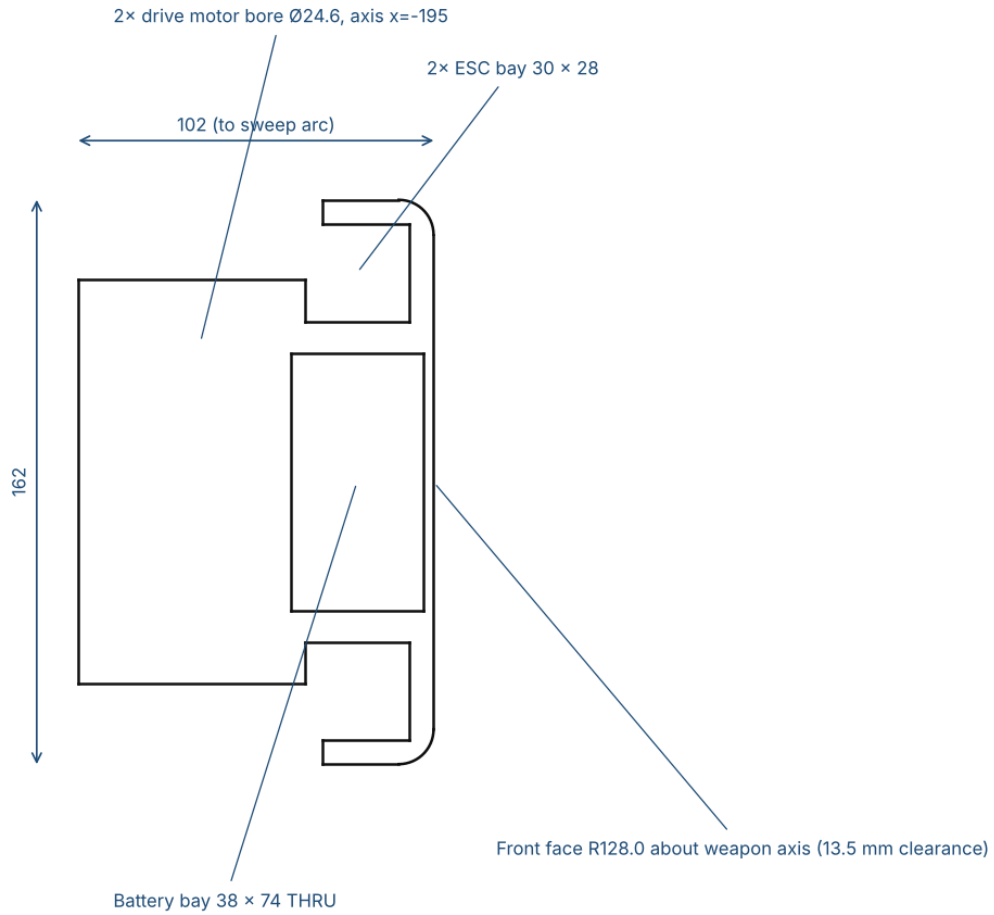


PART	DWG NO.	MATERIAL	THICKNESS	QTY	UNITS / SCALE
Bottom plate	**UMK2-003**	**Ti-6Al-4V (grade 5)**	**1.2 mm**	**1 (+1 spare)**	**mm / fit**

PROCESS: Laser or waterjet from bottom_plate.dxf. Deburr. Motor holes reamed for location.

NOTES: Tol ±0.15; corner radii R6; motor PCD true position Ø0.1 to datum A. Bottom plate also carries the fork (drill 2× Ø3.4 per fork drawing).

UNSKINNY MK II · REV A · 2026-11-07 · Drawn: Mahak · Master geometry: STEP/DXF in out/



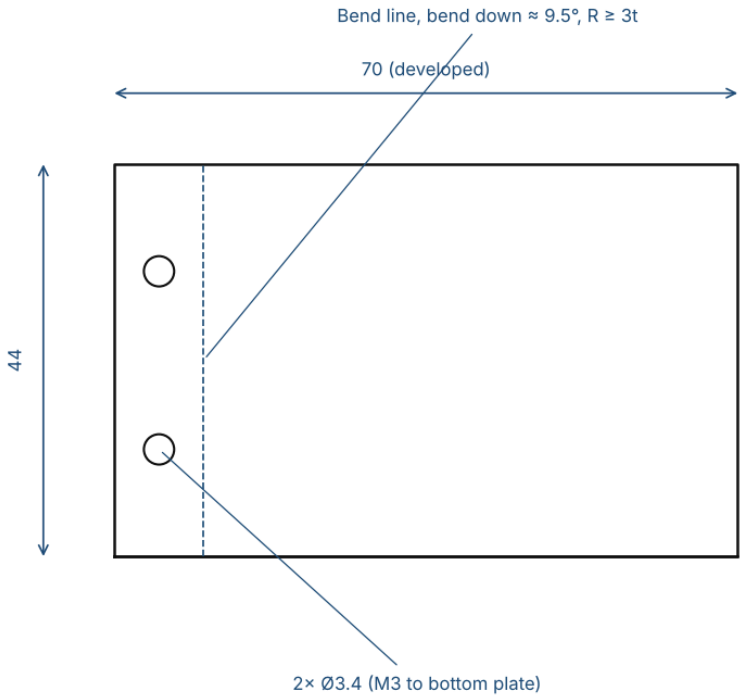
Height 40 mm (between plates). x positions measured from weapon axis (datum A).

PART	DWG NO.	MATERIAL	THICKNESS	QTY	UNITS / SCALE
Frame (rear pod)	**UMK2-004**	**TPU 95A**	**40 mm tall**	**1 (+1 spare)**	**mm / fit**

PROCESS: FDM print from frame.stl/frame.step: 0.2 mm layers, 5 walls, 45% gyroid, 0.4 nozzle, 230 °C, slow (30 mm/s).

NOTES: Install 12× M3 heat-set inserts (6 top, 6 bottom) at plate hole positions. Print flat. Check bores with a drive motor before final assembly.

UNSKINNY MK II · REV A · 2026-11-07 · Drawn: Mahak · Master geometry: STEP/DXF in out/



Tip must sit 0.5–1.0 mm above the floor with wheels loaded. Sharpen leading edge to 30°.

PART	DWG NO.	MATERIAL	THICKNESS	QTY	UNITS / SCALE
Front fork (wedgelet)	**UMK2-005**	**Ti-6Al-4V (or hardened spring steel)**	**1.5 mm**	**1 (+2 spares)**	**mm / fit**

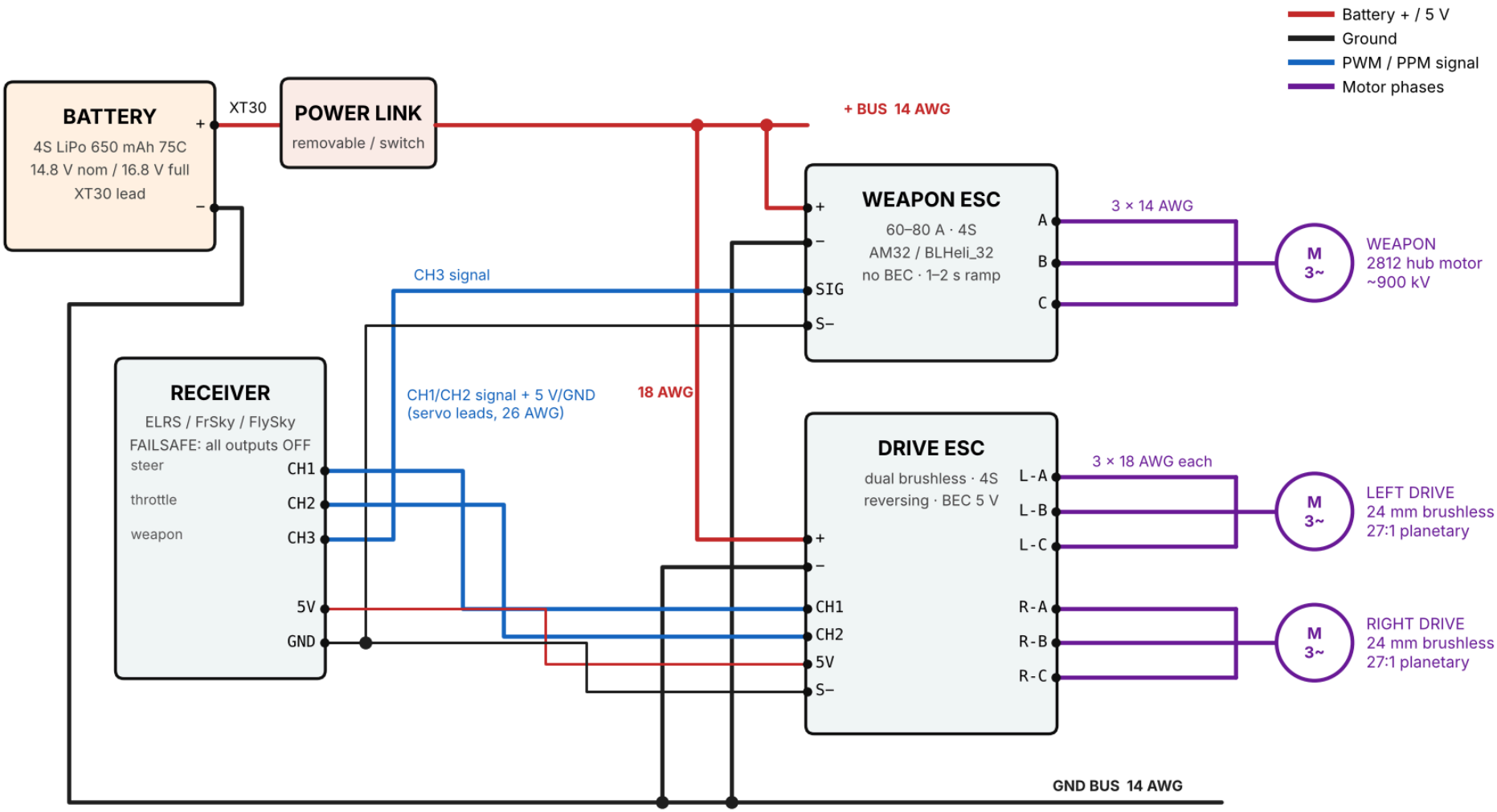
PROCESS: Laser cut flat, bend on brake.

NOTES: Fit-check height on a flat surface; shim with washers if needed.

UNSKINNY MK II · REV A · 2026-11-07 · Drawn: Mahak · Master geometry: STEP/DXF in out/

7 · Wiring

UMK2-E01 · Wiring diagram · Rev A



NOTES 1. Receiver is powered ONLY from the drive-ESC BEC; weapon ESC has no BEC (if yours has one, remove its red servo wire). 2. Swap any two phase wires to reverse a motor.
3. Arcade mixing on the transmitter OR the ESC, not both. 4. All joints soldered + heat-shrunk; strain-relief every lead; keep phase wires short. 5. Power link must be reachable with the robot fully assembled.

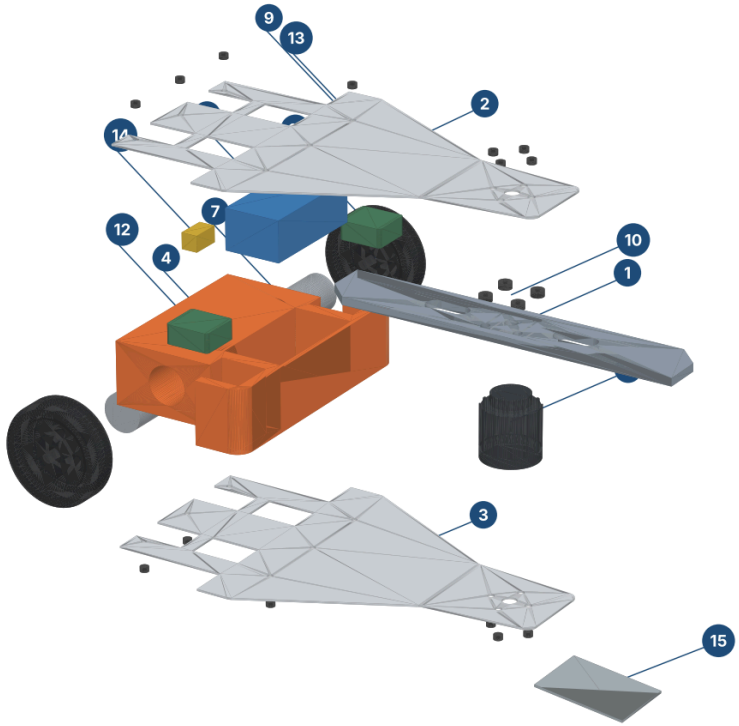
Wire list

From	To	Wire	Connector	Approx. length
Battery +/−	Power link	14 AWG silicone	XT30	60 mm
Power link	Weapon ESC +, Drive ESC +	14 AWG / 18 AWG	solder splice	80 / 60 mm
GND bus	Weapon ESC −, Drive ESC −	14 AWG / 18 AWG	solder splice	80 / 60 mm
Weapon ESC A/B/C	Hub motor	3 × 14 AWG	3.5 mm bullets	≤ 70 mm
Drive ESC L-A/B/C, R-A/B/C	Drive motors	6 × 18 AWG	2 mm bullets	≤ 50 mm
Receiver CH1/CH2/CH3	Drive ESC / Weapon ESC	26 AWG servo	JR/Dupont 3-pin	as needed

Receiver channels

Channel	Function	Goes to	Transmitter setting
CH1	Steering	Drive ESC CH1	Arcade mix (on TX or ESC, not both); rate 70% to start
CH2	Throttle	Drive ESC CH2	Centred = stop; both directions
CH3	Weapon	Weapon ESC SIG	On a switch or slider; low = off; never on a sprung stick
CH5 (optional)	Weapon brake / arm	Weapon ESC (if supported)	2-position switch

8 · Assembly



ITEM	QTY	DESCRIPTION
1	1	Weapon bar UMK2-001
2	1	Top plate UMK2-002
3	1	Bottom plate UMK2-003
4	1	Frame UMK2-004 (TPU)
5	2	Wheel + hub (L/R)
6	1	2812 hub motor
7	2	Drive motor (L/R)
8	1	4S 650 mAh LiPo
9	12+4	M3/M4 cap screws
10	4	4× M5×12 12.9 + Loctite 243
11	1	Weapon ESC
12	1	Drive ESC
13	1	Receiver
14	1	Power link
15	1	Fork UMK2-005

Assembly order: 4 → 7 → 5 → 3 → 15 → 6 → 1 + 10 → 11/12/13/14/8 → 2 + 9.
Threadlocker on all metal-metal threads.
Torques: M3 0.8 N·m (into inserts 0.5), M4 2.0 N·m, M5 4.5 N·m.

UMK2-A01 · Exploded assembly · Rev A · Unskinny Mk II

1. **Check the frame (4):** push a drive motor into each bore. It should be a firm press fit; ream with a heated rod if too tight. Install the 12 M3 heat-set inserts at 220 °C, flush with the surface.
2. **Drive (7, 5):** press the drive motors in, route their leads into the ESC bays, then fit the hubs and wheels. Spin by hand to check for rubbing.
3. **Bottom plate (3) and fork (15):** bolt the fork on, then bolt the bottom plate to the frame (M3, 0.5 N·m). Put it on a flat surface and check the fork tip sits 0.5–1.0 mm above it.
4. **Weapon motor (6):** bolt the stator flange to the bottom plate (M4 + Loctite, 2.0 N·m). It must spin freely.
5. **Bar (1, 10):** bolt the bar to the motor bell with 4× M5 12.9 + Loctite 243 at 4.5 N·m, tightening in a cross pattern. Check run-out at the tips is under 0.5 mm.
6. **Electronics (11–14, 8):** solder per §7, heat-shrink every joint, foam-wrap the ESCs, fit the receiver with its antenna away from the motor wires, mount the power link where you can reach it, strap the battery in.
7. **Top plate (2, 9):** fit it last, M4 into the motor (2.0 N·m) and M3 into the inserts (0.5 N·m).
8. **Weigh it.** It must be ≤ 1,361 g ready to fight. Record the actual weight of every part in `params.py`.

9 · Radio and speed-controller setup

1. Bind the receiver. Set failsafe to **"no pulses"**, or to all channels at stop and weapon at 0%.
2. **Weapon ESC** (BLHeli/AM32 configurator): direction counter-clockwise viewed from the top (§6 drawing); startup power low; **ramp-up 1–2 s**; low-voltage cutoff OFF (combat); timing medium; brake on stop optional (P2 upgrade).
3. Calibrate the weapon ESC throttle range to the transmitter (full → zero).
4. **Drive ESC:** reversing mode ON; confirm left and right directions and mixing. Swap two phase wires to reverse a motor.
5. **Failsafe test:** with the bar removed and the weapon running, switch the transmitter off. Every motor must stop within 1 s.

10 · Testing and quality checks

Test	Pass condition	Expected (simulated)
Weight	≤ 1,361 g	1,328 g
Bar balance	No visible wobble at 25% throttle	CoM < 0.05 mm
Failsafe	All motors stop < 1 s	—
Spin-up (test box)	90% speed in < 3 s	2.0 s
Weapon rpm (tachometer or telemetry)	≥ 9,000 rpm	~11,000 rpm
Drive speed	≥ 2 m/s across a 2.4 m box	2.7 m/s
3-minute run	ESCs/motors < 80 °C, pack > 3.6 V/cell at rest	7.4 of 7.7 Wh used
Inverted drive	Drives upside down	wheels 8.8 mm proud

Before every match

- ☐ Battery charged and balanced
- ☐ Bar screws torqued, teeth not cracked
- ☐ Plate screws all present
- ☐ Wheels secure, no TPU tears
- ☐ Fork height 0.5–1 mm
- ☐ Failsafe test passed
- ☐ Power link reachable, OFF
- ☐ Weight checked with fork fitted

11 · Spares and tools

Spares: bar ×1, teeth/inserts, top + bottom plate, frame ×1, forks ×2, drive motor ×1, weapon ESC, drive ESC, receiver, wheels ×2, batteries ×2, full screw set, heat-set inserts.

Tools: hex drivers 2/2.5/3/4 mm, torque driver (0.5–5 N·m), Loctite 243, soldering iron (60 W+), heat gun, multimeter, LiPo charger + fireproof bag, digital scale (1 g), tachometer, zip ties, Kapton tape, files/deburr tool.

12 · Safety

This robot stores about 940 J in a steel bar spinning at 130 m/s at the tips. Only spin the weapon inside a closed test box or the arena. Arm and disarm only with the weapon pointing away from everyone. Remove the power link whenever the robot is carried or worked on. Charge LiPos in a fireproof bag and never leave them charging unattended. Follow your event's safety rules and pass the tech inspection.